

Comparative Study of Lightweight Deep Learning Architectures for Responsible Edge AI Deployment in Semiconductor Defect Classification

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DECLARATION

I, **Prasad Malai (Roll No: 2023MCS320002]**), hereby declare that, this report entitled “**Comparative Study of Lightweight Deep Learning Architectures for Responsible Edge AI Deployment in Semiconductor Defect Classification**” submitted to Indian Institute of Information Technology Kottayam towards partial requirement of **Master of Technology in Computer Science and Engineering with Specialization in Big Data & Machine Learning** is an original work carried out by me under the supervision of **Dr. Ebin Deni Raj** and has not formed the basis for the award of any degree or diploma, in this or any other institution or university. I have sincerely tried to uphold the academic ethics and honesty. Whenever an external information or statement or result is used then, that have been duly acknowledged and cited.

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[**Prasad Malai**]

May 2026

CERTIFICATE

This is to certify that the work contained in this project report entitled **“Comparative Study of Lightweight Deep Learning Architectures for Responsible Edge AI Deployment in Semiconductor Defect Classification”** submitted by **Prasad Malai (Roll No: 2023MCS320002)** to Indian Institute of Information Technology Kottayam towards partial requirement of **Master of Technology in Computer Science and Engineering with Specialization in Big Data & Machine Learning** has been carried out by him under my supervision and that it has not been submitted elsewhere for the award of any degree.

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May 2026

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ABSTRACT

Semiconductor defect classification is crucial in ensuring product quality and yield in modern semiconductor manufacturing processes. This work explores the feasibility of deploying edge AI-based solutions for accurate and efficient defect classification, addressing the challenges of limited computational resources and real-time inference requirements. A comparative study of lightweight convolutional neural network architectures, including MobileNetV2, ShuffleNet, and ResNet18, is conducted for multi-class defect classification.

The models are trained and subsequently optimized for edge deployment using techniques such as model quantization and ONNX-based conversion, resulting in reduced latency and enhanced cross-platform portability. Model performance is evaluated in terms of classification accuracy, inference efficiency, and suitability for real-time edge deployment. To enhance model interpretability, multiple explainable artificial intelligence techniques are employed, including Gradient-weighted Class Activation Mapping, Integrated Gradients, and Occlusion Sensitivity, to identify the image regions that most strongly influence model predictions. Grad-CAM provides class-discriminative localization of salient defect areas, while Integrated Gradients and Occlusion Sensitivity offer complementary pixel-level and perturbation-based explanations of model behavior. Quantitative evaluation of the interpretability results is performed using metrics such as Mean Intersection over Union (IoU), Activation Ratio, and Average Drop, enabling systematic assessment of localization quality and the reliability of the highlighted defect regions.

Experimental results demonstrate that ResNet18 achieves superior localization performance due to its deeper architecture, while MobileNetV2 and ShuffleNet offer a favorable trade-off between computational efficiency and accuracy. The findings highlight the importance of balancing model complexity and interpretability for responsible edge AI deployment in semiconductor defect classification.

Keywords:

Edge Computing, Semiconductor Inspection, Scanning Electron Microscopy (SEM),
Raspberry Pi, Quantization, Explainable AI, Grad-CAM

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Chapter 1

Introduction

In recent years, the semiconductor industry has experienced significant growth driven by advancements in manufacturing technologies and the increasing demand for highly efficient, reliable, and high-performance integrated circuits. In modern semiconductor fabrication, defect detection and classification play a critical role in maintaining production efficiency and ensuring product reliability. Despite advancements in inspection systems (automatic and semi-automatic), the reduced size of integrated circuits and the increasingly complex nature of defect patterns continue to pose significant challenges to these approaches. Although deep learning techniques have demonstrated promising results in automated defect analysis, their deployment in real-world environments is often constrained by high computational requirements and limited interoperability across platforms. This has led to increasing interest in lightweight, edge-deployable solutions that can operate efficiently in resource-constrained and heterogeneous environments.

1.1 Background of Defect Classes

1.1.1 Semiconductor Manufacturing Overview

Semiconductor manufacturing is a highly complex, multi-stage process used to fabricate integrated circuits (ICs) on silicon wafers. The manufacturing process involves repeated cycles of lithography, deposition, etching, doping, and polishing, all performed with nanometer-level precision. Each wafer undergoes numerous processing steps in a clean room environment. Even minor variations in process conditions can introduce defects that adversely affect circuit functionality.

1.1.2 What are Wafer Defects and Why They Occur

Wafer defects refer to unintended irregularities or anomalies that appear either on the surface or within a semiconductor wafer during the fabrication process. These defects can originate from multiple sources within the manufacturing environment. For instance, contamination by microscopic particles in cleanroom conditions can lead to surface imperfections. Similarly, equipment-related issues, such as malfunctioning tools or improper calibration, may introduce inconsistencies during processing. Variations in critical process parameters, including temperature, pressure, and timing, can also affect the uniformity of the wafer. In addition, inherent material imperfections in silicon wafers can contribute to the formation of defects, impacting overall device quality and yield.

1.1.3 Types of Defects

- **Structural Defects:** Structural defects refer to physical damages that occur on the surface of a semiconductor wafer, including scratches, cracks, and breaks in the wafer structure. These defects are typically introduced due to

improper handling during fabrication or as a result of mechanical stress encountered in processing stages. Because they directly affect the physical integrity of the wafer, such defects are generally easy to detect using imaging systems. Furthermore, depending on their severity, structural defects can significantly degrade or even completely disrupt the functionality of the underlying circuitry.

- **Contamination Defects:** Contamination defects refer to the presence of unwanted foreign particles or residues on the surface of a semiconductor wafer. These may include dust particles, chemical residues, or metallic contaminants introduced during various stages of fabrication. Such defects are typically random in distribution and may be extremely small in size; however, even minute contamination can significantly affect device performance and yield. These defects are often introduced during cleaning processes or in pre and post-processing steps, where strict environmental and handling controls are essential to minimize their occurrence.
- **Pattern-Related Defects:** Pattern-related defects occur when the intended geometric layout of an integrated circuit is not accurately reproduced on the wafer, resulting in deviations from the designed pattern. Common examples include bridging, where unintended connections form between circuit lines; open circuits, which create breaks in expected conductive paths; misalignment of patterns; and the presence of missing or extra features. These defects typically arise during critical fabrication stages such as lithography or etching. Unlike structural defects, they may not always be easily identifiable through simple visual inspection, yet they have a direct and often severe impact on circuit behavior and overall device functionality.

1.1.4 Traditional Inspection approaches

- **Optical Inspection Method:** The optical inspection method is a widely used approach for detecting defects in semiconductor wafers, relying on high-resolution imaging systems to capture detailed wafer images during the manufacturing process. These images are then analyzed to identify irregularities such as scratches, contamination, or deviations in circuit patterns. Typically, the inspection process involves capturing images of the wafer, comparing them with reference patterns or adjacent dies, and flagging any deviations as potential defects. Despite its effectiveness, this method has certain limitations. It is often associated with high implementation costs and can be sensitive to noise in the captured images. Additionally, detecting defects in increasingly complex patterns, especially in advanced technology nodes, remains challenging. The method also generally requires further post-processing and interpretation to accurately classify and understand the detected defects.
- **Manual Classification:** Manual classification involves human experts visually inspecting wafer maps or microscope images to identify and categorize defect types. This approach relies heavily on the expertise and experience of the individual performing the analysis. While it can be effective for small-scale evaluations or complex cases requiring domain knowledge, it has several limitations. The process is inherently time-consuming and becomes impractical for high-volume manufacturing environments. Additionally, it is subjective in nature, leading to potential inconsistencies in defect identification and labeling across different engineers. Such variability can affect the reliability and reproducibility of the results, making manual classification less suitable for large-scale, standardized inspection systems.

Due to the limitations of traditional inspection methods, there has been a significant shift toward automated defect classification using machine learning and deep learning techniques. These approaches aim to improve accuracy, scalability, and real-time decision-making in semiconductor manufacturing environments.

1.2 Importance of defect classification

Defect classification is a key component of semiconductor manufacturing, as it influences product quality, overall yield, and the effectiveness of automated production processes. As device geometries shrink and fabrication complexity increases, accurate and efficient defect analysis becomes essential.

1.2.1 Quality Improvement

Defect classification plays a critical role in improving the overall quality of semiconductor manufacturing by ensuring that only wafers meeting stringent standards proceed to packaging and deployment. By accurately identifying and categorizing defects such as scratches, contamination, and pattern irregularities, manufacturers can detect faulty dies at an early stage in the production cycle. This early detection helps prevent defective chips from reaching end users, thereby reducing potential failures and associated costs. Moreover, it ensures consistent reliability in high-performance applications, including automotive, medical, and aerospace systems. The importance of defect classification is particularly pronounced in safety-critical domains, such as automotive embedded systems, where even minor defects can lead to significant system failures and compromise operational safety.

1.2.2 Yield Enhancement

Yield, defined as the percentage of functional chips produced from a semiconductor wafer, is a key metric in manufacturing efficiency. Defect classification plays a significant role in improving yield by enabling the identification of recurring defect patterns associated with specific process steps. This facilitates effective root cause analysis, allowing issues such as lithography errors or etching inconsistencies to be diagnosed and corrected. As a result, manufacturers can reduce scrap rates and minimize rework costs. Furthermore, by continuously analyzing defect trends over time, fabrication parameters can be optimized to enhance process stability and increase the number of usable chips per wafer, ultimately leading to improved productivity and higher profitability.

1.2.3 Process Automation

Traditional defect inspection approaches in semiconductor manufacturing have historically relied on manual review processes, which present several limitations. These methods are often time-intensive, prone to human error, and difficult to scale in high-volume production environments. In contrast, automated defect classification systems, particularly those leveraging deep learning techniques, offer significant advantages. They enable real-time detection and classification of wafer defects, ensure consistent and objective decision-making, and can be seamlessly integrated into modern smart manufacturing frameworks aligned with Industry 4.0 principles. By reducing reliance on human intervention, such automated systems enhance operational efficiency and support scalable, high-throughput fabrication processes.

1.2.4 Cost Reduction

Accurate defect classification contributes significantly to cost reduction in semiconductor manufacturing by minimizing various sources of operational inefficiency. By identifying defects at an early stage, it helps reduce material wastage, lowers the need for rework and repair, and prevents costly downtime caused by undetected process issues. Early detection ensures that defects do not propagate through subsequent stages of fabrication, thereby avoiding compounding losses. As a result, manufacturers can achieve more efficient resource utilization and substantial savings in overall production costs.

1.2.5 Data-Driven Manufacturing

Accurate defect classification contributes significantly to cost reduction in semiconductor manufacturing by minimizing various sources of operational inefficiency. By identifying defects at an early stage, it helps reduce material wastage, lowers the need for rework and repair, and prevents costly downtime caused by undetected process issues. Early detection ensures that defects do not propagate through subsequent stages of fabrication, thereby avoiding compounding losses. As a result, manufacturers can achieve more efficient resource utilization and substantial savings in overall production costs.

1.3 Limitations and Motivation

1.3.1 Limitation with current approach

The current approaches to defect classification face several important limitations that impact their effectiveness in practical manufacturing environments. High-

performing deep learning models, such as InceptionNet (GoogLeNet) and DenseNet, often require substantial computational resources, making them expensive to deploy and maintain. Many implementations also depend on cloud-based processing, which can introduce latency and bandwidth constraints, limiting their suitability for real-time defect detection in high-throughput production settings. Additionally, the black-box nature of these models reduces transparency and trust, particularly in critical manufacturing decision-making scenarios. Another challenge is their limited ability to generalize across different fabrication environments, as performance may degrade when applied to new or varying conditions. These models also rely heavily on large volumes of labeled data, which can be difficult and costly to obtain. At the same time, traditional inspection methods, while more interpretable, often lack robustness, whereas deep learning models, despite their accuracy, may not always be efficient for edge deployment.

1.3.2 Motivation for Using Edge AI

- Enables real-time defect classification directly on inspection machines
- Reduces latency by avoiding cloud communication
- Minimizes network bandwidth usage (important for high-resolution wafer images)
- Improve data privacy and security (sensitive manufacturing data stays on-site)
- Supports scalable deployment across multiple inspection units in fabs
- Allows integration with Industry 4.0 smart manufacturing pipelines

In conclusion, as semiconductor manufacturing moves towards fully automated and intelligent systems, the role of efficient and explainable defect classification becomes increasingly important. The convergence of Edge AI and interpretable deep learning offers a promising direction to achieve real-time, reliable, and scalable solutions. This work is driven by the vision of enabling practical, trustworthy edge AI systems that can be seamlessly integrated into next-generation smart manufacturing environments.

Chapter 2

Literature Overview

The semiconductor manufacturing industry has witnessed rapid advancements in device miniaturization, making defect detection and classification increasingly challenging. Early inspection systems relied on manual visual examination, which was feasible when defect sizes were relatively large. However, as feature sizes in semiconductor devices have scaled down to the nanometer range, defects have become more subtle and complex, necessitating the development of automated and highly accurate inspection systems. Advances in imaging technologies such as Scanning Electron Microscopy (SEM), which offers resolution in the range of 1–10 nm, have enabled the capture of high-quality wafer images, thereby facilitating the application of computational techniques for defect analysis.

2.1 Traditional Machine Learning Approaches

Initial efforts in automated defect classification employed traditional machine learning (ML) techniques such as Support Vector Machines (SVM), Decision Trees (DT), and K-Nearest Neighbors (KNN). These methods rely heavily on hand-

crafted feature extraction techniques, including texture, shape, and statistical descriptors. While these approaches demonstrated moderate success in controlled environments, their performance is highly dependent on the quality of feature engineering and lacks robustness when applied to complex and diverse defect patterns. Additionally, these methods struggle to generalize across varying imaging conditions and defect types, limiting their scalability in modern semiconductor manufacturing.[32]

2.2 Deep Learning-Based Approaches

To overcome the limitations of traditional ML methods, deep learning (DL), particularly Convolutional Neural Networks (CNNs), has been widely adopted for defect detection and classification. CNNs are capable of automatically learning hierarchical feature representations directly from raw image data, eliminating the need for manual feature extraction.[22][29]

Several studies have demonstrated the effectiveness of CNN-based architectures such as ResNet for wafer defect classification tasks.[9] These models achieve high accuracy in both binary and multiclass classification scenarios, often exceeding 95% performance. For instance, a hybrid framework combining a Binary Neural Network (BNN) for defect detection with a ResNet-based classifier for defect categorization has shown promising results in balancing detection speed and classification accuracy. However, such multi-stage architectures increase system complexity and may introduce latency during inference.

In addition to classification models, object detection frameworks such as YOLO have been extensively used for defect localization and classification in industrial applications[34], including printed circuit board (PCB) inspection [24]. These approaches enable real-time detection and accurate localization of defects such as

missing components and short circuits. Despite their advantages, YOLO-based models typically require large volumes of annotated data and may face challenges in detecting extremely small or low-contrast defects commonly found in semiconductor wafers.

2.2.1 Lightweight and Efficient Models

With the increasing demand for real-time inspection and deployment on resource-constrained systems, recent research has focused on developing lightweight and computationally efficient models.[10][9][3] Architectures such as EfficientNet and MobileNet have been explored to reduce model size and inference time while maintaining competitive accuracy.[13]

For example, an enhanced defect detection framework integrating EfficientNet-B0 with YOLOv8 has demonstrated significant improvements in detection performance, achieving a mean Average Precision (mAP@0.5) of 0.993 on the NEU-DET dataset. The use of Mobile Bottleneck Convolution (MBConv) blocks and multi-scale feature fusion enables efficient feature extraction and improved detection of defects with varying sizes. While such models are promising for industrial deployment, their evaluation is often limited to non-semiconductor datasets, raising concerns regarding their generalization to wafer defect scenarios.

2.2.2 Advanced Approaches: Multimodal and Transformer-Based Methods

Recent advancements in defect classification have explored multimodal learning and transformer-based architectures to further enhance performance[33]. Multimodal approaches integrate multiple data sources, such as SEM images and Energy-Dispersive X-ray (EDX) spectral data, to capture both spatial and com-

positional characteristics of defects[15]. These methods employ CNNs for image feature extraction and Multi-Layer Perceptrons (MLPs) for spectral data processing, followed by feature fusion using attention mechanisms. Such approaches have demonstrated improved classification accuracy and robustness compared to single-modality models. However, they introduce increased system complexity and require synchronized multimodal data acquisition.

Transformer-based models, including Vision Transformers (ViTs) and self-supervised frameworks such as DinoV2, have also gained attention for defect classification tasks [8]. These models are capable of capturing long-range dependencies and complex feature relationships, leading to strong performance even with limited labeled data. Experimental results indicate that transformer-based approaches can achieve classification accuracies exceeding 90% using transfer learning and semi-supervised techniques. Nevertheless, their high computational and memory requirements pose significant challenges for deployment in real-time and edge-based industrial environments.

2.2.3 Comparative Analysis of Existing Methods

2.3 Comparative Analysis of Existing Methods

A comparative analysis of existing approaches highlights the trade-offs between accuracy, computational efficiency, and deployment feasibility. Traditional ML methods are computationally efficient but lack robustness and scalability. CNN-based deep learning models achieve high accuracy but often require substantial computational resources. Lightweight architectures improve efficiency and are suitable for real-time applications; however, they may compromise performance when handling

complex defect patterns. Multimodal and transformer-based approaches further enhance accuracy and feature representation but introduce increased complexity and computational overhead.

2.4 Research gap

Key Research Gaps identified include:

2.4.1 Accuracy Priority Over Deployability

A significant limitation of existing approaches is the strong emphasis on maximizing classification accuracy, often at the expense of deployability in real-world manufacturing environments. Many high-performing models are built with a large number of parameters, requiring powerful GPU-based servers for training and inference. This leads to increased inference latency, which is not ideal for time-sensitive production lines. Additionally, such models are difficult to deploy directly on inspection machines due to hardware constraints, and their reliance on high-end computational resources results in increased implementation and maintenance costs. As a result, despite achieving high accuracy, these approaches may not be practical for efficient, real-time industrial deployment. [12][18]

2.4.2 Limited Edge AI Adoption

Most existing defect classification models are trained and evaluated in resource-rich environments such as cloud platforms, high-performance workstations, and experimental setups including Kaggle or laboratory systems. While these settings facilitate the development and benchmarking of complex models, they do not accurately represent real-world industrial deployment conditions. Consequently, only

a limited number of studies focus on practical implementation aspects such as on-device inference[33], optimization for low-power processors, or deployment on industrial embedded boards. Furthermore, the challenge of enabling real-time inspection at local nodes within manufacturing environments remains largely underexplored. This gap highlights the need for solutions that are not only accurate but also efficient and suitable for edge-based deployment.

2.4.3 Very few studies address actual industry needs:

Despite significant advancements in defect classification, relatively few studies adequately address practical industry requirements, particularly in terms of model explainability and trust. Many deep learning systems operate as black boxes, providing high accuracy without offering clear insights into their decision-making processes. However, in semiconductor fabrication, engineers require more transparent and interpretable outputs to support critical decisions. For instance, it is important to understand why a particular wafer was rejected, which specific defect regions influenced the prediction, and whether the model is focusing on actual defects such as scratches or merely reacting to noise or irrelevant patterns. The lack of such interpretability limits the adoption of deep learning models in real-world industrial settings, where reliability and trust are essential.

Chapter 3

Problem Statement & Research overview

3.1 Problem Statement

Semiconductor fabrication facilities generate massive volumes of high-resolution wafer and die inspection images, where early and accurate identification of defects is essential for maintaining yield, reliability, and cost efficiency. While automated optical inspection systems are widely used for defect detection, the subsequent defect classification stage frequently relies on centralized cloud servers or manual expert analysis. This dependency creates several bottlenecks, such as communication latency, network bandwidth limitations, privacy concerns related to sensitive manufacturing data, and increased operational costs. Additionally, these issues hinder real-time inspection in semiconductor manufacturing lines. Hence, there is a critical need for an Edge-AI-enabled defect classification framework that can execute directly on low-power edge hardware with limited memory and computational resources. The proposed system should deliver high classification performance with

reduced latency and optimized resource utilization, enabling practical and scalable deployment in semiconductor fabrication environments.

3.2 Objective

3.2.1 Performance Benchmark Report

To systematically compare selected deep learning models for semiconductor defect classification using multiple evaluation metrics such as Accuracy, Precision, Recall, F1-score, inference latency, and memory usage. In many research works, models are compared only using classification accuracy. However, for real-world semiconductor manufacturing systems, high accuracy alone is insufficient. A model must also be fast, lightweight, and efficient.

Therefore, this objective focuses on creating a benchmark report that evaluates each candidate model (such as MobileNet, ResNet, ShuffleNet, etc.) using:

- Accuracy: Overall percentage of correctly classified defect images.
- Precision: How many predicted defects are correctly identified.
- Recall: Ability to detect all actual defects.
- F1-score: Balanced metric combining precision and recall.
- Inference Latency: Time required to classify one image.
- Memory Consumption: RAM/storage required during deployment.
- Model Size: Disk space occupied by trained model.
- Computational Complexity: FLOPs or parameter count.

Expected Outcome:

A comparative table ranking all models and identifying the best trade-off between performance and efficiency.

3.2.2 Edge Deployment Feasibility Report

To assess the practical suitability of the trained models for deployment on low-resource edge devices, experiments were conducted on platforms such as the Raspberry Pi 4 using quantitative performance metrics. While many deep learning models demonstrate strong performance on GPUs or cloud-based systems, their effectiveness often diminishes on edge hardware due to inherent constraints such as limited CPU capability, restricted memory availability, thermal limitations, power constraints, and slower storage access. To address these challenges and evaluate real-world feasibility, several key measurements were recorded after deploying the models on the Raspberry Pi 4. These include the actual inference time per image, CPU utilization, RAM consumption, and frames per second (FPS). Additionally, the impact of optimization techniques such as quantization and pruning was analyzed by measuring any resulting drop in classification accuracy. This evaluation provides a comprehensive understanding of the trade-offs between performance and efficiency in edge deployment scenarios.

3.2.3 Responsible AI Assessment

To ensure that the selected AI model is transparent, unbiased, fair, and trustworthy prior to deployment in semiconductor manufacturing, a comprehensive evaluation of its reliability and interpretability is essential. As industrial adoption of AI systems continues to expand, there is an increasing need for models that not only achieve high classification accuracy but also provide explainable and dependable

decision-making. In wafer defect inspection, misclassification can result in serious consequences such as yield loss, unnecessary rejection of functional wafers, failure to detect critical defects, and substantial financial impact.

To address these concerns, this study incorporates multiple dimensions of responsible AI assessment. Explainability analysis is performed using Gradient-weighted Class Activation Mapping, Integrated Gradients, and Occlusion Sensitivity to identify the image regions that most strongly influence model predictions. Grad-CAM provides class-discriminative localization of salient defect regions, while Integrated Gradients offers pixel-level attribution by estimating the contribution of individual input features. Occlusion Sensitivity complements these methods by quantifying the effect of masking localized image regions on prediction confidence, thereby validating whether the highlighted areas are truly relevant to the classification decision.

In addition to interpretability, bias analysis is conducted across different defect categories to examine whether the model exhibits disproportionate performance variations among classes. Fairness assessment is further performed to ensure that all defect types are treated consistently without undue preference toward any specific category. Together, these analyses provide a more comprehensive understanding of model behavior and strengthen confidence in the reliability, transparency, and practical suitability of the proposed framework for semiconductor defect classification.

3.2.4 Design Guidelines for Edge vs Cloud Deployment

To develop recommendations that identify appropriate scenarios for deploying semiconductor defect classification on edge platforms versus cloud-based systems.

This study also aims to develop practical recommendations for identifying suitable scenarios in which semiconductor defect classification should be deployed on edge platforms versus cloud-based systems. Edge deployment is particularly advantageous in situations where real-time response and low latency are critical, such as in high-throughput production lines requiring immediate accept or reject decisions. It is also preferred in environments with limited or unreliable internet connectivity, where local processing ensures uninterrupted operation. Additionally, edge-based solutions help address data privacy concerns and are well-suited for continuous, high-volume inspection tasks, enabling real-time quality monitoring directly on-site. In contrast, cloud-based deployment is more appropriate for computationally intensive tasks such as large-scale model training, storage of extensive datasets, and historical analytics. It also facilitates cross-facility monitoring and periodic model retraining using aggregated global data. Typical use cases include long-term yield trend analysis and centralized model updates, where latency is less critical but computational capacity and scalability are essential.

Chapter 4

System Methodology

This chapter presents the methodology adopted for the development of the proposed Edge-AI-based semiconductor defect classification system. The objective of the system is to accurately classify semiconductor defects while satisfying practical deployment constraints such as low latency, reduced memory usage, and computational efficiency on resource-constrained edge devices.

The overall approach includes preparing the dataset, applying image preprocessing and augmentation techniques, selecting suitable lightweight models, performing transfer learning-based training, evaluating performance, optimizing the models, and assessing their feasibility for deployment on edge devices. In addition, explainability and responsible AI assessment techniques are incorporated to ensure reliable and trustworthy model predictions.

4.1 Workplan of proposed system

The overall work plan of the proposed system is shown in Figure 4.1. The process starts with the collection of semiconductor defect datasets, which are then sub-

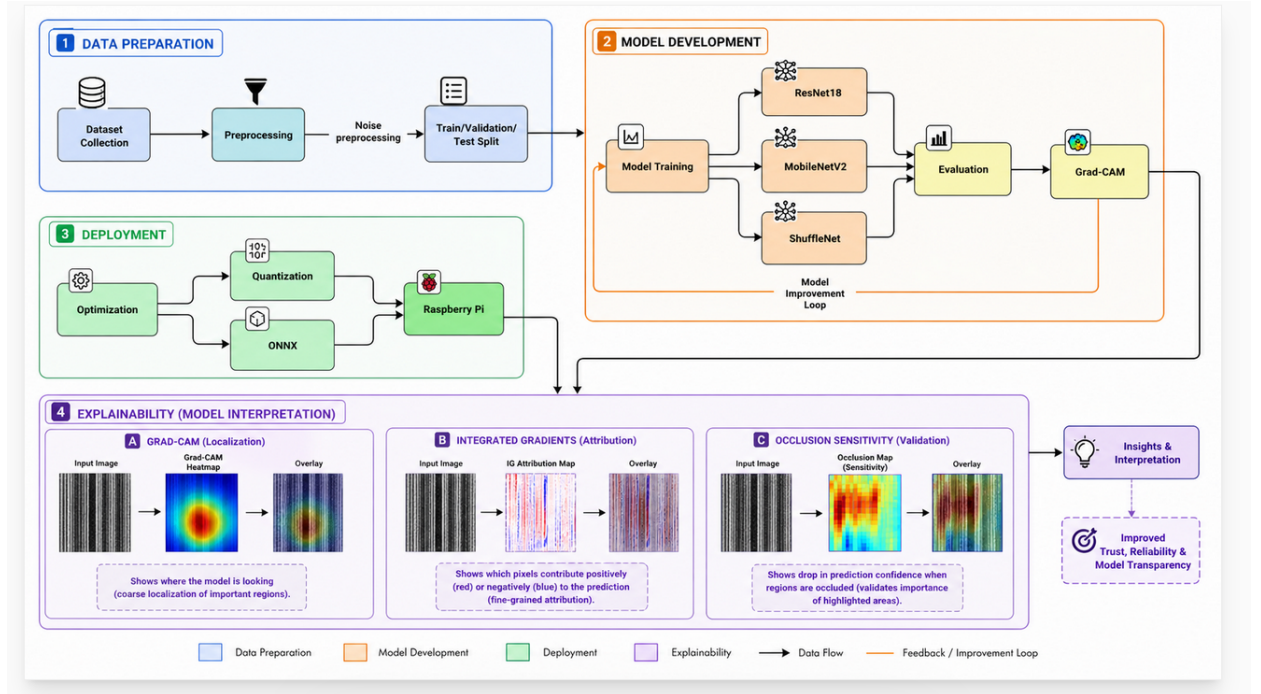


Figure 4.1: Overall Work Plan of the Proposed Semiconductor Defect Classification System

jected to preprocessing steps such as resizing and normalization to ensure uniform input quality. After preprocessing, the dataset is divided into training, validation, and testing sets using a stratified splitting approach to maintain balanced class distribution. This phase took approximately 1.5 weeks.

In the next stage, different deep learning models, including MobilenetV2, ShuffleNetV2 and ResNet18, are developed and trained using the prepared dataset. Their performance is analyzed using evaluation metrics such as accuracy and F1-score. To improve model interpretability, Gradient-weighted Class Activation Mapping, Integrated Gradients, and Occlusion Sensitivity are employed to identify and analyze the image regions that most strongly influence the model's predictions. This stage required approximately two weeks.

Following this, model optimization is carried out using quantization techniques to reduce computational complexity and make the model suitable for efficient edge deployment. Finally, the optimized model is deployed in an edge computing environment to enable real-time semiconductor defect classification. This final phase also took approximately 2 weeks.

The subsequent sections provide a detailed discussion of each phase involved in the proposed workflow.

4.2 System Architecture

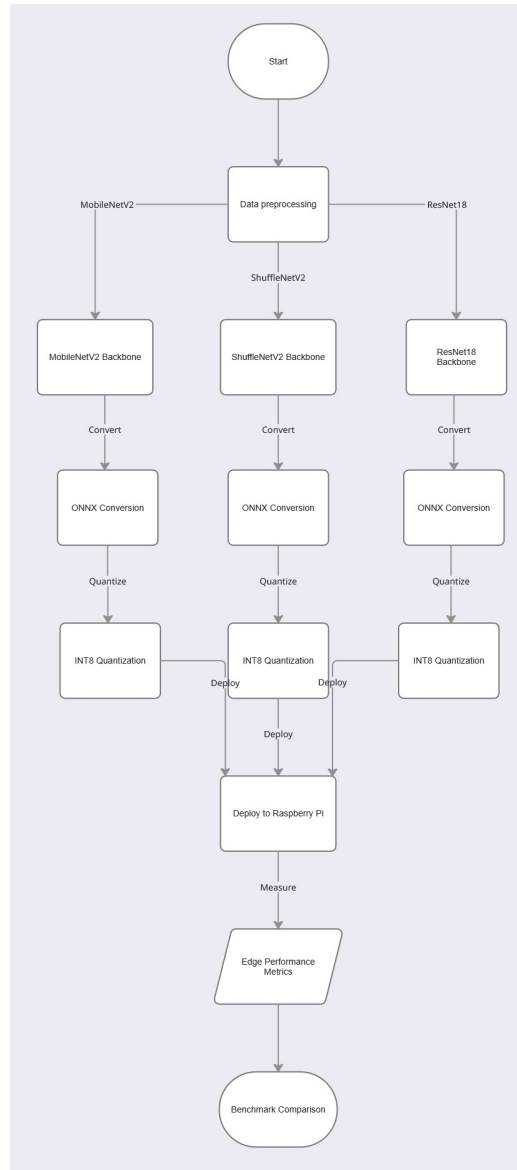


Figure 4.2: Overall System Architecture

The proposed system architecture follows a structured pipeline for semiconductor defect classification, beginning with data acquisition and preprocessing, where raw wafer images are normalized and prepared for model input. The preprocessed data is then fed into multiple deep learning models, including lightweight architectures and deeper networks, enabling comparative analysis. Each model performs feature extraction followed by classification to identify defect categories. The outputs from these models are evaluated and compared to determine performance in terms of accuracy and efficiency. To enhance model interpretability, an explainability module is integrated, which highlights the regions of the input image that influence the model’s decision. Finally, the system performs comparative analysis across all models, facilitating the selection of an optimal approach that balances accuracy and computational efficiency for real-time deployment in resource-constrained environments. A detailed information on each phase can be observed in below subsequent sections

4.3 Data collection and Preparation

The performance of any deep learning-based defect classification system is highly dependent on the quality, diversity, and representativeness of the training dataset. In semiconductor manufacturing, obtaining large-scale labeled defect datasets is a major challenge due to intellectual property restrictions, confidentiality agreements, limited public release of fab inspection data, and the high cost of expert annotation. As a result, publicly available semiconductor image datasets often contain limited defect categories, low sample counts, or incomplete metadata.

To address these practical constraints, a hybrid dataset generation strategy was adopted in this work by combining publicly available semiconductor defect images with synthetically generated defect samples. This approach enabled the

creation of a balanced and sufficiently diverse dataset suitable for training and evaluating lightweight deep learning models under realistic research constraints.

4.3.1 Data Collection Approach

Since access to proprietary industrial semiconductor datasets was not available, the dataset used in this research was developed from two complementary sources:

Public Semiconductor Defect Datasets

Publicly available semiconductor defect datasets were explored to obtain realistic inspection images representing defects observed during wafer or die analysis. These datasets provide valuable insights into real-world manufacturing conditions, capturing defect textures, process-induced noise, and variations that are difficult to replicate synthetically. By leveraging such datasets, the study aims to incorporate authentic inspection characteristics into the model development process. Specifically, these datasets help capture key aspects such as realistic image properties, practical background noise, irregular and complex defect boundaries, diverse real-world imaging conditions, and variations in illumination and contrast. This ensures that the trained models are exposed to representative data, thereby improving their robustness and generalization capability in practical deployment scenarios.

Synthetic Defect Image Generation

Due to the limited availability of publicly labeled data, a synthetic dataset was generated to complement the existing real-world samples. Artificial defect patterns were designed to closely resemble semiconductor layout anomalies by incorporating geometric structures, shape distortions, line interruptions, crack-like formations,

and irregular conductive paths. This approach enabled the creation of diverse and representative defect scenarios that are often underrepresented in public datasets. The use of synthetic data helped address several key challenges, including limited sample size, absence of certain defect categories, class imbalance, and insufficient variation in pattern scale and orientation. As a result, the augmented dataset improved the robustness and generalization capability of the trained models.

4.3.2 Defect Classes Considered

The chosen dataset focuses on defects relevant to lithography, interconnect structures, and layout inspection processes.

LER (Line Edge Roughness)

This defect is characterized by irregular or rough edges along patterned semiconductor features, typically arising from variations in lithography or etching processes. Excessive edge roughness can negatively impact device performance and dimensional accuracy.

Crack

Crack defects represent visible fractures or discontinuities in material layers, lines, or structural regions. These defects can reduce reliability and may lead to open circuit failure.

Malformed Via

This defect occurs when inter-layer connection holes are improperly formed, misaligned, partially developed, or deformed. Such irregularities can disrupt electrical connectivity between layers.

Open

Open defects represent broken conductive paths where current flow is interrupted due to missing metal connection or structural discontinuity.

Short

Short defects occur when two conductive regions become unintentionally connected, resulting in undesired electrical conduction.

These five classes were selected as they represent both geometric and electrical failure modes commonly encountered in semiconductor inspection systems.



Figure 4.3: Semiconductor defects

4.3.3 Data Challenges

Before model development, several challenges were encountered, including:

Class Imbalance

Certain defect classes have more /less samples compared to others. For instance, crack and malformed via defects were less frequent than open or short patterns. This imbalance can bias the model toward majority classes, potentially affecting its generalization performance.

Limited Access to Real Industrial Data

Access to real industrial defect data in semiconductor manufacturing is often limited, as such data may expose proprietary circuit layouts, process technologies, yield-related information, or other customer-sensitive details. As a result, large-scale industrial datasets are generally not publicly available for research purposes. In addition to data scarcity, another major challenge arises from the high variability in defect appearance. Defects can differ significantly in terms of shape, orientation, size, intensity, contrast, position, and background texture, making accurate classification more complex. This variability increases the difficulty of developing robust models and necessitates the use of effective preprocessing and data augmentation techniques to ensure reliable performance across diverse defect scenarios.

4.3.4 Data Pre-processing

Prior to model training, all collected images were transformed into a standardized format to ensure compatibility with convolutional neural network architectures.

Image Resizing

Images originally collected from different sources contained varying resolutions. To maintain compatibility with pretrained lightweight CNN architectures, all images were resized to:

224×224 pixels

This size was selected because it is the standard input dimension for many ImageNet-pretrained models such as MobileNet, ResNet, and EfficientNet.

It has few benefits as:

- Uniform tensor dimension

- Fast batch processing
- Lower memory footprint
- Compatibility with transfer learning approach

Normalization

Pixel intensity values were normalized using standard ImageNet mean and standard deviation values:

- Mean = [0.485, 0.456, 0.406]
- Std = [0.229, 0.224, 0.225]

Normalization improves numerical stability and accelerates convergence during training. It is especially useful when using transfer learning from ImageNet-pretrained networks.

4.3.5 Data Augmentation

To improve generalization and reduce overfitting, data augmentation techniques were applied during training. Augmentation artificially increases dataset diversity by generating transformed versions of existing images while preserving defect identity.

The following transformations were used:

Rotation

Images were randomly rotated within a specified angle range. This helps the model learn rotational invariance since defects may appear at different orientations.

Flipping

Horizontal and vertical flipping were used where physically meaningful. This improves robustness to mirrored pattern appearances.

Scaling / Zooming

Random scaling allowed the model to recognize defects appearing at different sizes or magnification levels.

Benefits of Augmentation

Data augmentation offers several important benefits in the context of semiconductor defect classification by enhancing the diversity and quality of the training data. It improves the model's ability to generalize to unseen data by exposing it to a wider range of variations during training. Additionally, augmentation helps reduce overfitting by preventing the model from relying too heavily on specific patterns present in limited datasets. It also supports better learning of minority classes by artificially increasing their representation, thereby addressing class imbalance. Furthermore, augmentation increases the model's robustness to real-world inspection variability, such as changes in orientation, scale, and imaging conditions, ultimately leading to more reliable performance in practical deployment scenarios.

4.3.6 Dataset Description

The complete dataset was divided into three mutually exclusive subsets for unbiased training and evaluation.

Training Set

Used to train the deep learning model and update network weights.

Validation Set

Used during training to tune hyperparameters, monitor overfitting, and select the best-performing model checkpoint.

Test Set

Used only after training completion to measure final generalization performance on unseen data.

```
Classes: ['LER', 'crack', 'malformed_via', 'open', 'short']  
Number of classes: 5
```

```
==== Dataset Summary ====  
Training samples : 1032  
Validation samples : 122  
Testing samples : 136  
Total samples : 1290
```

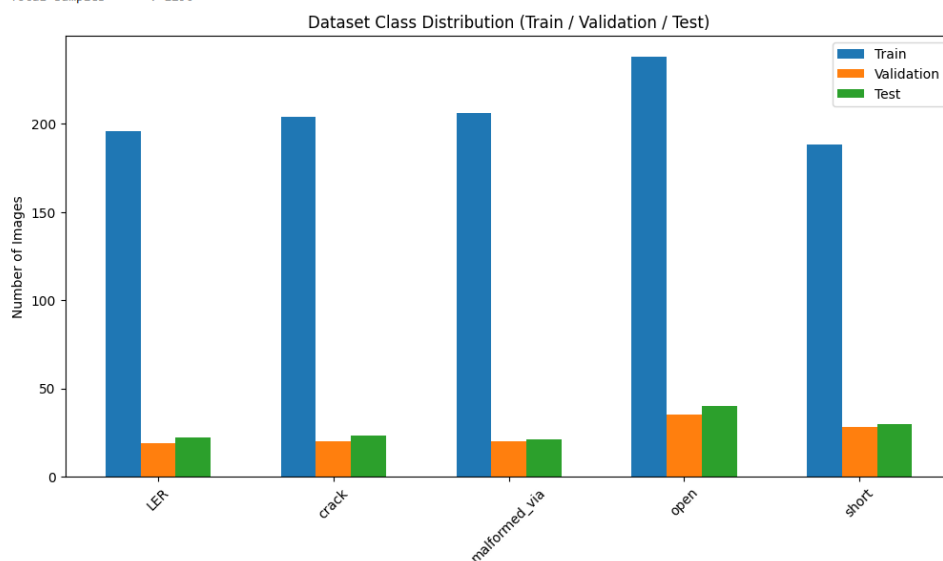


Figure 4.4: Dataset distribution

A split ratio used for this work:-

Training Set:	75%
Validation Set:	15%
Test Set:	15%

Owing to the limited availability of real industrial semiconductor defect datasets, a hybrid data strategy was employed, combining both public and synthetically generated samples. Five key defect classes were considered in this study. The dataset was subjected to preprocessing, normalization, augmentation, and a structured train-validation-test split to support robust model training and reliable evaluation of the proposed Edge-AI framework.

4.4 Model Architecture and Training Methodology

Selecting an appropriate deep learning CNN architecture is a critical step in the development of an Edge-AI-based semiconductor defect classification system. In practical deployment environments, the model must not only achieve high classification accuracy but also satisfy hardware constraints such as low memory usage, reduced computational complexity, and fast inference speed.

Therefore, this work focuses on lightweight and edge-deployable convolutional neural network (CNN) architectures that are suitable for deployment on embedded platforms such as Raspberry Pi and similar low-power devices. To enable a meaningful comparison across different network characteristics, three convolutional neural network architectures were selected: MobileNetV2, ShuffleNet, and ResNet18. These models were chosen to represent varying trade-offs between computational efficiency and representational capacity.

- MobileNetV2 – designed for efficient execution on resource-constrained de-

vices by employing depthwise separable convolutions

- ShuffleNet – focuses on reducing computational cost through channel shuffling and grouped operations, making it suitable for low-power hardware
- ResNet18 – utilizes residual connections to facilitate deeper feature learning while maintaining a relatively compact structure, serving as a reference baseline

All models were trained and evaluated using the same semiconductor defect dataset under uniform experimental settings to ensure a fair and consistent comparison.

4.4.1 Edge Deployable Model Selection Criteria

The selected models were chosen based on the following criteria:

Low Parameter Count

Smaller models require less storage and memory, making them suitable for embedded systems.

Reduced Computational Cost

Lower FLOPs (floating point operations) enable faster inference.

Transfer Learning Availability

Pretrained ImageNet weights help improve performance with limited semiconductor datasets.

Real-Time Inference Capability

Models should support low-latency classification for inline semiconductor inspection systems.

4.4.2 Model 1: MobileNetV2

Architecture Overview

The architecture overview of MobileNetV2 highlights its design as a compact and efficient convolutional neural network optimized for edge and embedded platforms. The model incorporates key architectural innovations such as depth-wise separable convolutions, inverted residual blocks, and linear bottleneck layers. These components work together to significantly reduce computational complexity and memory requirements while maintaining competitive classification accuracy. As a result, MobileNetV2 is well-suited for deployment in resource-constrained environments where efficiency and performance must be carefully balanced.

Why MobilnetV2 is selected

MobileNetV2 is particularly well-suited for semiconductor edge inspection tasks due to its balanced design that prioritizes both efficiency and performance. It offers a low memory footprint, enabling deployment on resource-constrained devices, along with fast inference speed, which is essential for real-time inspection scenarios. Despite its lightweight nature, the model maintains strong classification performance, making it reliable for defect detection. Additionally, its architecture is optimized for efficient deployment on platforms such as the Raspberry Pi and other embedded systems, further supporting its applicability in practical edge-based inspection environments.

Training Methodology

The pre-trained ImageNet MobileNetV2 model was utilized as the base network to leverage transfer learning for defect classification. The original fully connected classification layer was replaced with a custom output layer consisting of five neurons, corresponding to the target defect classes: LER, Crack, Malformed Via, Open, and Short. To effectively adapt the model to the semiconductor domain, a staged fine-tuning strategy was employed. Initially, the early layers of the network were frozen to retain generic feature representations, while the final layers were re-trained using the semiconductor defect dataset. Subsequently, partial fine-tuning was performed by unfreezing selected deeper layers, allowing the model to better capture domain-specific features and improve overall classification performance.

4.4.3 Model 2: ShuffleNet

ShuffleNet is an ultra-lightweight convolutional neural network architecture specifically designed for deployment on highly resource-constrained devices. Its efficiency is achieved through key design components such as pointwise group convolution, channel shuffle operations, and effective feature reuse mechanisms. These architectural innovations significantly reduce computational complexity and memory requirements while preserving the model’s ability to extract meaningful features. As a result, ShuffleNet provides a practical balance between efficiency and performance, making it suitable for real-time applications in edge-based environments.

Why Shufflenet is selected

ShuffleNet is particularly well-suited for edge benchmarking due to its highly efficient architecture designed for resource-constrained environments. It features a very low parameter count, which reduces memory requirements and computa-

tional overhead. This enables faster inference on CPU-based systems, making it ideal for deployment on devices without dedicated GPUs. As a result, ShuffleNet is well adapted for real-time embedded applications, where low latency and efficient processing are critical for continuous and high-throughput operations.

Training Methodology

The pretrained ShuffleNet model was initialized using weights learned from the ImageNet dataset to leverage transfer learning. The original classification layer was replaced with a custom output layer consisting of five neurons corresponding to the target defect classes. A structured fine-tuning approach was adopted to adapt the model to the semiconductor defect classification task. Initially, the base layers retained their pretrained weights to preserve general feature representations, while the final classification layer was retrained using the target dataset. Subsequently, the entire network was fine-tuned using a low learning rate, allowing gradual adaptation of deeper layers to domain-specific features while maintaining training stability and preventing overfitting.

4.4.4 Model 3: ResNet18

Architecture Overview

ResNet18 is a residual convolutional neural network containing 18 layers. It uses skip connections that help train deeper networks by avoiding vanishing gradients.

Why ResNet18 is selected

ResNet18 was selected as a reference model despite its relatively larger size compared to lightweight architectures such as MobileNetV2 and ShuffleNet. Its inclusion is motivated by its strong feature extraction capability and stable training

behavior, enabled by residual connections that facilitate efficient gradient flow. These characteristics allow ResNet18 to achieve consistent and reliable performance across a variety of classification tasks. In this study, it serves as a benchmark model to evaluate the trade-offs between accuracy and computational efficiency, providing a meaningful baseline for comparison with more resource-efficient architectures.

Training Methodology

A pretrained ResNet18 model was employed as the base architecture to leverage transfer learning for defect classification. The original fully connected layer was replaced with a custom classifier consisting of five output neurons corresponding to the target defect classes. Following this modification, the entire network was fine-tuned on the semiconductor dataset to adapt the learned features to the specific domain. A lower learning rate was used during training to ensure stable convergence and to prevent large weight updates that could disrupt the pretrained feature representations.

4.5 Training Methodology

Deep learning models trained in research environments are usually developed on high-performance systems equipped with GPUs, large memory, and powerful processors. However, in practical semiconductor manufacturing environments, deploying these models directly on edge hardware such as Raspberry Pi, industrial embedded controllers, or low-power CPUs presents several challenges.

Edge devices operate under limited computational resources, restricted memory, lower power budgets, and real-time response requirements. Therefore, a trained model must be optimized before deployment so that it can perform ef-

efficient inference without significantly compromising classification accuracy.

This section explains the need for edge optimization techniques such as quantization and the importance of ONNX as a deployment framework.

4.5.1 Why Edge Optimization is Required

Although a model may achieve strong performance during training, deploying it directly on edge hardware without optimization can lead to several practical challenges. These include slow inference speed, high memory consumption, prolonged startup or loading time, excessive CPU utilization, and thermal throttling due to sustained processing loads. Additionally, unoptimized models may consume more power and fail to meet the real-time requirements expected in industrial environments. In semiconductor production lines, such inefficiencies can disrupt the inspection process and reduce overall throughput. Therefore, model optimization is essential to ensure efficient, reliable, and real-time performance when deployed on resource-constrained edge devices.

4.5.2 Quantization

Quantization is a model optimization technique that reduces the numerical precision of weights and activations, typically resulting in lower computational complexity and smaller model size. In the context of edge deployment, quantization plays a crucial role in improving inference efficiency by enabling faster computations on standard processors using reduced-precision representations. It also reduces runtime memory requirements, as fewer bits are needed to store model parameters, making it well-suited for devices with limited resources. Additionally, quantized models are more compact, simplifying storage, transfer, and deployment on edge platforms. This is particularly important for devices such as the Raspberry

Pi, which have constrained RAM and CPU capabilities. By applying quantization, models can be executed more efficiently without relying on high-performance GPU hardware, making them practical for real-time, resource-limited environments.

4.5.3 Why ONNX is Needed

ONNX (Open Neural Network Exchange) is a standardized format for representing machine learning models in a framework-independent manner, enabling models trained in platforms such as PyTorch or TensorFlow to be exported into a portable and optimized representation. This portability facilitates seamless deployment across different environments, particularly in edge-based systems. In contrast, directly deploying models using their original frameworks can introduce several challenges, including the need for large software installations, additional runtime dependencies, increased memory overhead, and slower CPU-based inference. Furthermore, such deployments often involve complex setup procedures, making them less suitable for embedded and resource-constrained devices. As a result, utilizing ONNX helps streamline deployment and improves efficiency in edge computing scenarios.

4.5.4 Quantization + ONNX Together

An efficient deployment pipeline can be established by sequentially performing model training, conversion to the ONNX format, application of quantization techniques, and final deployment on the target edge device. This structured workflow enables the transformation of a high-performance model into a form that is suitable for resource-constrained environments. By following this approach, the resulting model benefits from a reduced memory footprint, faster inference speed, and lower runtime RAM usage. Consequently, it becomes feasible to achieve practical real-

time deployment on edge platforms, making the system suitable for industrial applications such as semiconductor defect inspection.

Edge optimization is essential because models trained in high-performance environments are often too heavy for direct embedded deployment. Techniques such as quantization reduce model size, memory usage, and inference latency. ONNX further improves portability and execution efficiency by providing a lightweight standardized deployment format. Together, these techniques make real-time semiconductor defect classification feasible on practical edge devices.

4.6 Explainable AI-Based Analysis of the Proposed Semiconductor Defect Classification System

In this project, Grad-CAM was primarily used as a post-training validation and interpretability tool to examine how the trained deep learning models arrived at their classification decisions. Since the proposed system is intended for semiconductor defect inspection, it is important to ensure that the model predicts a defect class based on the actual defect region rather than irrelevant image patterns.

Therefore, Grad-CAM was integrated into the evaluation stage of the project and applied after model training and testing.

4.6.1 Application of Explainability Techniques in this Study

After training the defect classification models MobileNetV2, ShuffleNet, and ResNet18 on the semiconductor defect dataset, multiple explainability techniques were employed to examine the interpretability of the learned predictions using test images. For each input sample, the trained model first produced a predicted defect category. Based on this prediction, three complementary explainability methods Gradient weighted Class Activation Mapping, Integrated Gradients, and Occlusion Sensitivity were applied to understand the image regions that contributed most strongly to the classification decision.

Grad-CAM was computed using the final convolutional layer of the network to generate a class-discriminative heatmap highlighting the spatial regions that most influenced the model prediction. Integrated Gradients was used to estimate the contribution of individual input pixels by measuring their attribution toward the predicted class. Occlusion Sensitivity was applied by systematically masking localized image patches and observing the corresponding change in prediction confidence, thereby identifying regions that were most critical to the classification outcome.

The outputs of these methods were visualized as attribution maps and overlaid on the original semiconductor defect images. The highlighted regions were then qualitatively compared with the actual defect locations to assess whether the models were focusing on physically meaningful defect structures rather than on irrelevant background patterns. This analysis was carried out for both correctly classified and misclassified samples. Correctly classified samples helped verify that the models had learned meaningful defect-related representations, while misclassified samples provided insight into possible causes of incorrect predictions.

The main objective of applying these explainability techniques in this study was to validate that the trained models were making decisions based on relevant semiconductor defect characteristics rather than on spurious artifacts or noise. The combined use of Grad-CAM, Integrated Gradients, and Occlusion Sensitivity provided complementary perspectives on model behavior, thereby improving confidence in the transparency and reliability of the proposed defect classification framework.

4.6.2 Use of Explainability Techniques Across Defect Categories

Explainability analysis using Gradient-weighted Class Activation Mapping, Integrated Gradients, and Occlusion Sensitivity was conducted for all five defect classes to examine whether the models attended to defect-relevant regions during classification.

LER (Line Edge Roughness)

The explainability maps were used to verify whether the models concentrated on irregular line boundaries, edge fluctuations, and variations in line smoothness.

Crack

The attribution maps were examined to confirm whether the models focused on visible crack or fracture-like regions associated with this defect class.

Malformed Via

The visual explanations were used to inspect whether the models localized the circular via region and the surrounding geometric distortion.

Open

The methods were used to verify whether the models attended to broken conductive paths, missing links, or local discontinuities.

Short

The attribution maps were used to confirm whether the models focused on bridging regions or unintended conductive connections that distinguish short-circuit defects.

4.6.3 Interpretation of Model Failures through Explainability Visualization

Explainability analysis using Gradient-weighted Class Activation Mapping, Integrated Gradients, and Occlusion Sensitivity was further employed to interpret model failures by examining incorrectly classified test samples. For each misclassified instance, these techniques helped reveal whether the model focused on irrelevant image regions, failed to capture subtle defect characteristics, confused visually similar defect patterns, or was influenced by background noise and image artifacts.

This analysis provided valuable insight into the limitations of the trained models and highlighted possible areas for improvement, including stronger data augmentation, increased dataset diversity, and architectural refinement. In several misclassified cases, the attribution maps indicated that prediction errors often occurred when the defect region was weakly localized, when the discriminative features were visually subtle, or when surrounding texture patterns shared similarities with other defect categories.

In this study, the explainability methods were not used solely as post-training visualization tools but also as practical evaluation mechanisms for comprehensively

assessing model behavior. They enabled class-wise attention analysis, comparison across different architectures, and detailed investigation of failure cases, thereby supporting more informed model selection for edge deployment.

By verifying that model decisions were primarily driven by actual defect regions rather than irrelevant image features, this interpretability-driven analysis improved confidence in the reliability, transparency, and practical suitability of the proposed semiconductor defect classification framework for industrial semiconductor manufacturing applications.

Chapter 5

Result analysis and discussion

5.1 Experimental setup

All models were trained using the PyTorch framework on a GPU-enabled environment. MobileNetV2 and ShuffleNet were trained for 20 epochs, while ResNet18 was trained for 15 epochs due to faster convergence. For optimization, MobileNetV2 and ResNet18 were trained using the Adam optimizer, while ShuffleNet employed AdamW to enhance generalization performance. Learning rates were set to 0.001 for MobileNetV2 and ShuffleNet, and 1e-5 for ResNet18 to ensure stable training. All models were trained with a batch size of 32 and an input image resolution of 224×224 . Cross-entropy loss was used as the optimization objective. Data augmentation techniques such as horizontal flipping and rotation were applied to improve model robustness.

Table 5.1: Training Parameters for MobileNetV2

Parameter	Value
Model	MobileNetV2
Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	20
Input Size	224×224
Number of Classes	5
Loss Function	Cross-Entropy Loss
Data Augmentation	Horizontal Flip, Rotation (10°), Normalization

Parameter	Value
Model	ShuffleNet
Optimizer	AdamW
Learning Rate	0.001
Batch Size	32
Epochs	20
Input Size	224×224
Number of Classes	5
Loss Function	Cross-Entropy Loss
Scheduler	ReduceLROnPlateau (patience = 3)
Data Augmentation	Resize, Grayscale $\rightarrow$ RGB, Random transforms

Table 5.2: Training Parameters for ShuffleNet

Parameter	Value
Model	ResNet18
Optimizer	Adam
Learning Rate	1e-5
Batch Size	32
Epochs	15
Input Size	224×224
Number of Classes	5
Loss Function	Cross-Entropy Loss
Data Augmentation	Normalization only with flip or rotation

Table 5.3: Training Parameters for ResNet18

5.2 Training Performance analysis

5.2.1 Training overview for Mobilnet V2

```

Validation Accuracy: 75.00%
F1 Score: 0.7474
      precision    recall  f1-score   support

    LER           0.64      0.70      0.67         10
   crack           0.67      0.73      0.70         11
malformed_via       0.67      0.80      0.73         10
      open           0.82      0.88      0.85         26
      short          0.85      0.58      0.69         19

   accuracy                   0.75         76
  macro avg           0.73      0.74      0.73         76
 weighted avg           0.76      0.75      0.75         76

Confusion Matrix:
[[ 7  0  0  2  1]
 [ 1  8  2  0  0]
 [ 1  1  8  0  0]
 [ 0  0  2 23  1]
 [ 2  3  0  3 11]]

```

Figure 5.1: MobilnetV2 Training Overview

Observation

An accuracy of approximately 75% is considered acceptable in this context, given the use of MobileNetV2, which prioritizes computational efficiency over representational complexity for edge deployment. The application of INT8 quantization further reduces numerical precision, contributing to a slight drop in accuracy. Considering the inherent complexity of SEM images and the limited computational capabilities of the Raspberry Pi platform, this result reflects a practical balance between predictive performance and real-time execution constraints.

5.2.2 Training overview for Shufflenet

```
Best Validation Accuracy: 0.868421052631579
Test Accuracy: 0.7901234567901234

Classification Report:
              precision    recall  f1-score   support

     LER          0.92         0.92         0.92         12
    crack          0.86         1.00         0.92         12
malformed_via      0.90         0.82         0.86         11
      open          0.70         0.76         0.73         25
      short         0.72         0.62         0.67         21

   accuracy          0.79         0.79         0.79         81
  macro avg          0.82         0.82         0.82         81
 weighted avg          0.79         0.79         0.79         81


Confusion Matrix:
[[11  0  0  0  1]
 [ 0 12  0  0  0]
 [ 0  1  9  0  1]
 [ 1  1  1 19  3]
 [ 0  0  0  8 13]]
F1 Score: 0.7873
```

Figure 5.2: ShuffleNet Training Overview

Observation

The ShuffleNet model attained an accuracy of approximately 76%, along with an F1-score of 0.75, indicating a balanced performance across the evaluated classes.

This is expected given its lightweight architecture, which prioritizes computational efficiency using grouped and depthwise convolutions. The confusion matrix indicates that some defect classes are misclassified due to visual similarity and limited feature representation. However, the model offers significant advantages in terms of low latency and small model size, making it highly suitable for deployment on resource-constrained edge devices like Raspberry Pi

5.2.3 Training overview for Resnet18

Accuracy: 0.9091				
F1 Score: 0.9076				
	precision	recall	f1-score	support
LER	0.96	0.96	0.96	24
crack	0.95	1.00	0.97	35
malformed_via	0.92	0.96	0.94	25
open	0.88	0.88	0.88	40
short	0.85	0.77	0.81	30
accuracy			0.91	154
macro avg	0.91	0.91	0.91	154
weighted avg	0.91	0.91	0.91	154

Figure 5.3: ResNet18 Training Overview

Observation

Among the evaluated models, ResNet18 achieved the highest classification accuracy of over 90%, which can be attributed not only to its deeper architecture and residual learning capability but also to effective hyperparameter tuning. Specifically, optimization of parameters such as learning rate, batch size, number of epochs, and regularization techniques contributed significantly to improved convergence and generalization performance. The residual connections in ResNet18 further facilitate stable gradient propagation, enabling the network to learn more discriminative features for complex defect patterns.

However, this improved performance comes at the cost of increased computational complexity and higher inference latency. As a result, despite achieving superior accuracy, ResNet18 is less suitable for real-time deployment on resource-constrained edge devices. Therefore, in this work, ResNet18 is considered as a benchmark model for performance comparison, while lightweight architectures are preferred for practical edge-based implementation due to their favorable trade-off between accuracy and efficiency.

5.3 Edge inference result over Raspberry Pi4

5.3.1 Edge inference result for MobilNetV2

```
(edge_env) prasad@prasad:~/Project $ python3 Raspberry_pi_inference.py
2026-03-29 14:35:59.247381071 [W:onnxruntime:Default, device_discovery.cc:325 DiscoverDevicesForPlatform] GPU device discovery failed: device_discovery.cc:92 ReadFileContents Failed to open file: "/sys/class/drm/card1/device/vendor"

===== EDGE BENCHMARK MobilnetV2=====
Input: input
Prediction: Crack (Class 1)
Latency: 243.13 ms
Throughput: 4.11 FPS
Average CPU Usage: 49.53%
RAM Before: 80.52 MB
RAM After : 80.53 MB
RAM Increase: 0.01 MB
(edge_env) prasad@prasad:~/Project $ |
```

Figure 5.4: Edge inference result for MobilNetV2

5.3.2 Edge inference result for ShuffleNet

```
(edge_env) prasad@prasad:~/Project $ python3 Raspberry_pi_inference.py
2026-03-29 14:31:25.532191757 [W:onnxruntime:Default, device_discovery.cc:325 DiscoverDevicesForPlatform] GPU device discovery failed: device_discovery.cc:92 ReadFileContents Failed to open file: "/sys/class/drm/card1/device/vendor"

===== EDGE BENCHMARK Shufflenet=====
Input: input
Prediction: Crack (Class 1)
Latency: 78.88 ms
Throughput: 12.81 FPS
Average CPU Usage: 50.02%
RAM Before: 64.10 MB
RAM After : 64.11 MB
RAM Increase: 0.01 MB
(edge_env) prasad@prasad:~/Project $ |
```

Figure 5.5: Edge inference result for ShuffleNet

5.3.3 Edge inference result for ResNet18

```
(edge_env) prasad@prasad:~/Project $ python3 Raspberry_pi_inference.py
2026-03-29 14:26:06.295630966 [W:onnxruntime:Default, device_discovery.cc:325 DiscoverDevicesForPlatform] GPU device discovery failed: device_discovery.cc:92 ReadFileContents Failed to open file: "/sys/class/drm/card1/device/vendor"

===== EDGE BENCHMARK =====
Input: input
Prediction: Crack (Class 1)
Latency: 352.20 ms
Throughput: 2.84 FPS
Average CPU Usage: 50.08%
RAM Before: 80.75 MB
RAM After : 80.76 MB
RAM Increase: 0.01 MB
(edge_env) prasad@prasad:~/Project $
```

Figure 5.6: Edge inference result for ResNet18

Classification Performance					
Sl.No	Model	Overall Accuracy	F1 Score	Model Size in MB After Quantization	Latency in ms (Kaggle)
1	MobileNetV2	0.7530	0.7400	2.23	99.53
2	ShuffleNet	0.7600	0.7594	1.30	28.19
3	ResNet18	0.9091	0.9076	10.70	128.71

Table 5.4: Classification Performance of Selected Models

Edge Inference Performance						
Sl.No	Model	Edge Suitability	Edge Latency (ms)	Throughput	Average CPU Usage	RAM Increase
1	MobileNetV2	Good	244.98	4.08	49.44	0.01
2	ShuffleNet	Excellent	78.08	12.81	50.02	0.01
3	ResNet18	Poor	352.20	2.84	50.08	0.01

Table 5.5: Edge Inference Performance of Selected Models

5.4 Explainable AI Based Qualitative and Quantitative Analysis

The model concentrated strongly on the fracture region, indicating that prediction was based on relevant structural discontinuity.

SL. No	Model Used	Mean IOU	Activation Ratio
1	ShuffleNet	0.2800	0.8140
2	MobileNetV2	0.3135	0.8153
3	ResNet18	0.349	0.8306

Table 5.6: Quantitative Grad-CAM Comparison of Selected Models

5.4.1 Grad-CAM visualization for Mobilenet V2

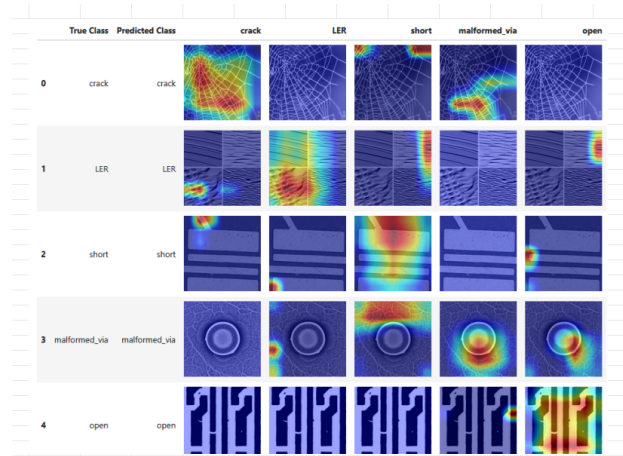


Figure 5.7: Grad-Cam over MobilNetV2

5.4.2 Grad-CAM visualization for ShuffleNet

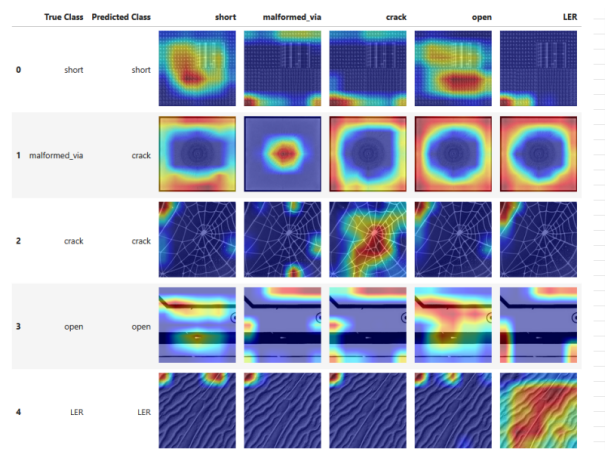


Figure 5.8: Grad-Cam over ShuffleNet

5.4.3 Grad-CAM visualization for ResNet18

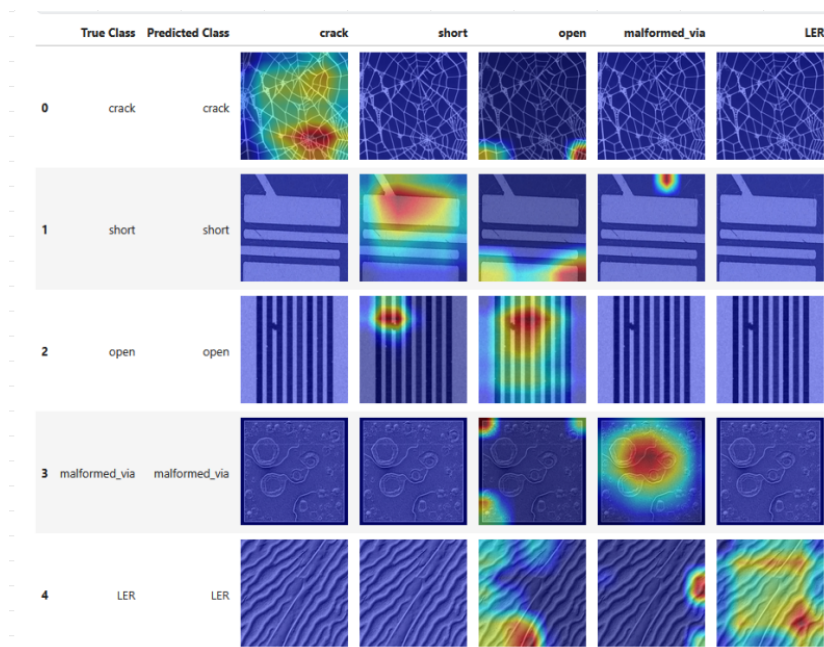


Figure 5.9: Grad-Cam over ResNet18

5.4.4 Integrated Gradients-Based Qualitative Interpretation of Model Predictions

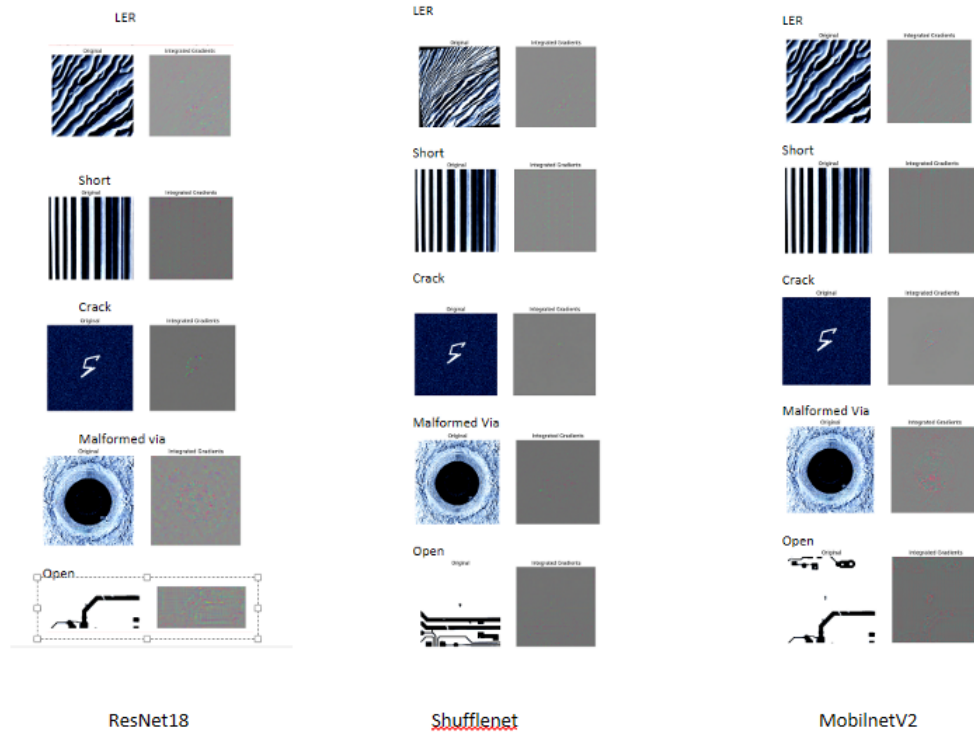


Figure 5.10: Qualitative Analysis Using Integrated Gradients

5.4.5 Occlusion-Based Qualitative Interpretation of Model Predictions

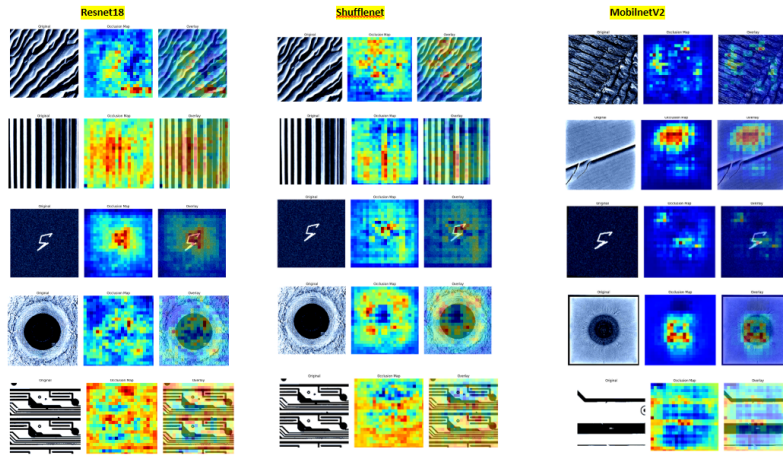


Figure 5.11: Occlusion-Based Interpretation

Table 5.7: Average Confidence Drop for Occlusion-Based Validation

SL. No	Model Used	Average Drop
1	ShuffleNet	0.1220
2	MobileNetV2	0.3149
3	ResNet18	0.1535

Table 5.8: Model-wise Interpretability Insights from Explainability Experiments

Technique	ResNet18	ShuffleNet	MobileNetV2
Grad-CAM	Produced relatively concentrated attention over defect regions	Attention aligned with defect areas but appeared slightly more diffuse	Highlighted relevant defect regions with moderate localization consistency
Integrated Gradients	Showed compact attribution around important defect structures	Displayed useful but comparatively scattered pixel-level attribution	Exhibited stronger localized attribution for several defect classes
Occlusion Sensitivity	Moderate confidence drop (0.1535), indicating balanced dependence on localized cues	Lowest confidence drop (0.1220), suggesting more distributed feature usage	Highest confidence drop (0.3149), indicating stronger reliance on localized defect regions

5.5 Observations & Reasoning:

5.5.1 Correlation Between Classification Performance and Localization

A strong correlation was observed between classification performance and localization capability across the evaluated models. Models achieving higher classification metrics, such as accuracy and F1-score (for example, ResNet18), also demonstrated superior localization performance, reflected in higher Mean IoU values and better alignment in Grad-CAM activation maps. In contrast, lower-performing models,

such as ShuffleNet, exhibited both reduced classification accuracy and weaker localization, with less focused activation regions. This relationship suggests that models capable of learning more discriminative and representative features not only perform better in classification tasks but also more accurately identify relevant spatial regions associated with defects. The Grad-CAM visualizations further support this observation by showing that correct predictions are typically associated with well-focused and meaningful activation patterns, reinforcing the link between classification effectiveness and spatial attention.

5.5.2 Integrated Gradients-Based Qualitative Interpretation of Model Predictions

A qualitative examination of the Integrated Gradients visualizations indicates that all three architectures—ResNet18, ShuffleNet, and MobileNetV2 primarily assign importance to localized image regions rather than to the entire wafer surface. For the LER and Short categories, the attribution responses are concentrated along the line structures visible in the original samples. This behavior suggests that the models are extracting prediction relevant information from edge irregularities, spacing variations, and line continuity patterns, which are consistent with the visual characteristics of these defect classes.

For the Crack category, the attribution maps show a clearer concentration around the defect-shaped region, indicating that the classifiers are able to associate the prediction with the actual crack morphology instead of relying mainly on surrounding background texture. In the Malformed Via samples, noticeable attribution appears around the circular boundary and adjacent disturbed regions, implying that geometric distortion of the via contributes substantially to class discrimination. These observations support the view that the learned representa-

tions capture physically meaningful structural cues associated with wafer defect patterns.

A comparatively different trend is observed for the Open class. In several examples, the attribution response is weaker and more spatially dispersed than in the other categories. This suggests that the prediction for this class may not depend on a single dominant visual region but instead may involve a combination of subtle discontinuities and surrounding texture information. Such behavior can indicate that the Open defect is comparatively harder to localize precisely, even when classification performance remains acceptable.

When the three models are compared qualitatively, the overall attribution patterns remain broadly consistent across architectures, which provides additional confidence that the predictions are not entirely driven by arbitrary background artifacts. However, minor differences in attribution compactness are visible. In some cases, ResNet18 produces slightly more concentrated responses, whereas ShuffleNet and MobileNetV2 occasionally exhibit more diffuse activations. This variation may be related to differences in feature extraction depth and architectural complexity. Overall, the Integrated Gradients analysis indicates that the models predominantly rely on defect-relevant structural features, supporting the interpretability of the classification framework.

5.5.3 Occlusion-Based Qualitative Interpretation of Model Predictions

Considering the average confidence drop obtained from Occlusion Sensitivity, the quantitative results further support the qualitative observations. Among the evaluated models, ResNet18 shows an average drop of 0.1535, ShuffleNet shows 0.1220, and MobileNetV2 shows the highest average drop of 0.3149. Since the average drop

measures the reduction in prediction confidence after masking the most relevant regions, a larger value indicates stronger dependence on the highlighted defect-related areas.

These results suggest that although all three architectures focus on meaningful defect regions, the degree of reliance differs across models. MobileNetV2 exhibits the strongest sensitivity to occlusion, indicating that its predictions depend more heavily on localized discriminative regions identified in the occlusion maps. ResNet18 shows moderate sensitivity, implying a balanced use of localized defect cues along with broader contextual information. In contrast, ShuffleNet records the lowest average drop, suggesting comparatively lower dependence on the masked high-importance regions and a more distributed use of spatial features during classification.

Overall, the quantitative occlusion results are consistent with the visual interpretation of the heatmaps. The relatively higher confidence reduction observed for MobileNetV2 indicates stronger localization of defect-relevant evidence, whereas the lower reductions for ResNet18 and ShuffleNet suggest that these models may additionally exploit surrounding texture and contextual information. Therefore, the combined qualitative and quantitative analysis confirms that the defect classification models learn meaningful structural representations, while differing in how strongly they rely on localized defect regions.

5.5.4 ResNet18 – High Accuracy, Strong Localization, Higher Cost

ResNet18 demonstrates high classification performance along with strong localization capability, albeit at a relatively higher computational cost. The model achieves superior accuracy by learning highly discriminative features, which also

enhances its ability to focus on relevant spatial regions within defect images. This is evident from Grad-CAM visualizations, which show precise and concentrated attention on actual defect areas. However, these performance benefits come with increased model size and comparatively higher inference time, making it less efficient for resource-constrained edge deployment. The effectiveness of ResNet18 can be attributed to its residual connections, which enable deeper feature learning without degradation and facilitate stable convergence during training. As a result, the model is capable of capturing both low-level texture details and high-level semantic features, leading to highly discriminative and spatially aligned representations that support accurate defect classification and localization.

5.5.5 MobileNetV2 – Balanced Trade-off

The model demonstrates slightly lower classification performance compared to ResNet18, with reduced accuracy and F1-score, along with a moderate Mean IoU, indicating acceptable but not optimal localization capability. Grad-CAM visualizations reveal that the model generally focuses on the correct defect regions; however, the attention is somewhat diffused, suggesting less precise spatial localization. Despite this, the model offers significant advantages in terms of efficiency, including a much smaller size and faster inference speed, making it well-suited for edge deployment. These characteristics arise from the use of depthwise separable convolutions, which effectively reduce computational complexity and memory requirements. However, this compression can lead to a partial loss of spatial detail, resulting in only moderate localization precision compared to deeper architectures.

5.5.6 ShuffleNet – Efficient but Limited Learning Capacity

The model exhibits the lowest classification performance among the evaluated architectures, with reduced accuracy, F1-score, and Mean IoU, indicating weaker localization capability. Grad-CAM visualizations further reveal that the activation maps are often diffused and occasionally misaligned with the actual defect regions, suggesting limited spatial focus. Despite these limitations, the model offers the smallest size and fastest inference speed, making it highly efficient for deployment in resource-constrained environments. These characteristics stem from its architectural design, which employs channel shuffling and grouped convolutions to reduce computational complexity. However, this efficiency comes at the cost of limited feature interaction and reduced representational capacity, which negatively impacts the model’s ability to capture fine-grained defect details and achieve precise localization.

5.5.7 Edge Deployment Trade-off Analysis

A comparative analysis of the three models reveals clear trade-offs between accuracy, localization capability, and computational efficiency. ResNet18 achieves the best classification accuracy and strongest localization performance; however, it incurs higher inference latency, making it less suitable for strict real-time edge constraints. MobileNetV2 provides a balanced trade-off between performance and speed, offering good accuracy with efficient inference, making it more practical for edge deployment scenarios. In contrast, ShuffleNet is the fastest and most lightweight model, but it exhibits the lowest reliability in terms of classification accuracy and localization precision. From a deployment perspective, edge systems require low latency and minimal memory usage to operate effectively in real-time

environments. However, insufficient localization accuracy, as observed through Grad-CAM analysis, can lead to incorrect inspection decisions, highlighting the importance of balancing efficiency with interpretability and reliability in industrial applications.

5.5.8 Grad-CAM Analysis for Correct and Incorrect Predictions

An analysis of Grad-CAM visualizations across different models reveals a clear relationship between model performance and attention behavior. High-performing models consistently exhibit strong activation within actual defect regions, with minimal interference from background areas, indicating effective learning of relevant spatial features. In contrast, moderately performing models tend to show shifted or dispersed activation patterns, where attention is either partially misaligned with defect regions or becomes fragmented across the image. This behavior suggests confusion arising from visually similar textures and limited spatial resolution in feature representation. Overall, Grad-CAM analysis confirms that misclassifications are strongly associated with incorrect or unfocused spatial attention, highlighting the critical role of interpretability techniques in evaluating model reliability and ensuring trustworthy defect detection in industrial applications.

Final comparative insight

The final comparative analysis highlights that high classification accuracy alone is not always sufficient for deployment in real-world semiconductor inspection systems. Similarly, models optimized primarily for speed and efficiency are not necessarily reliable in terms of prediction quality and localization performance.

Therefore, deployment decisions must consider a combination of factors, including classification accuracy, localization capability, inference time, and model size. Among the evaluated models, ResNet18 delivers superior classification accuracy and strong localization performance, making it suitable as a high-performance benchmark. However, MobileNetV2 provides a more practical balance between accuracy and computational efficiency, making it better suited for edge deployment scenarios. In contrast, ShuffleNet, despite its excellent efficiency and low resource consumption, may not be ideal for critical defect inspection tasks due to its limited localization precision and reduced representational capacity.

Chapter 6

Future work

One important extension of the present research is the development of a comprehensive deployment guideline framework for semiconductor defect classification systems operating in both edge and cloud environments. In the next phase of this work, a detailed guideline document will be prepared to support deployment decisions based on application requirements.

The proposed work will compare edge and cloud deployment scenarios using critical factors such as inference memory constraints, latency, computational resource availability, scalability, privacy considerations, operational cost, and maintenance complexity. This will assist semiconductor manufacturers in selecting the most suitable deployment architecture for specific inspection workflows. For example, latency-sensitive inline inspection tasks may benefit from edge deployment, whereas large-scale historical analytics and centralized model retraining may be more suitable for cloud environments. The future study will also investigate hybrid edge-cloud architectures combining real-time local inference with centralized intelligence.

Chapter 7

Conclusion

This thesis presented an Edge-AI-based framework for semiconductor defect classification aimed at overcoming the limitations of centralized inspection systems such as latency, bandwidth dependency, and higher infrastructure cost. A hybrid dataset comprising public and synthetic defect images was prepared to address limited access to industrial semiconductor data. Five important defect categories, namely LER, Crack, Malformed Via, Open, and Short, were considered.

Three lightweight deep learning architectures were trained and evaluated using standard classification metrics along with deployment oriented measures such as latency and memory usage. The experimental results demonstrated that lightweight models can achieve competitive classification performance while remaining suitable for resource constrained edge platforms.

Exporting the trained models to the ONNX format enabled smoother execution across different hardware platforms and reduced dependency on the original training framework. This contributed to more efficient inference, particularly in edge-based environments. In addition, Grad-CAM visualization was used to examine model behavior, confirming that the networks primarily attended to defect-specific

regions, which improves confidence in the reliability of the predictions.

Among the evaluated lightweight architectures, MobileNetV2 demonstrated the most balanced performance in terms of memory usage, inference speed, and overall suitability for edge deployment.

Overall, this research demonstrates that Edge-AI can play a significant role in next-generation semiconductor manufacturing by enabling faster, scalable, and intelligent defect analysis. Future work may extend this study toward hybrid edge-cloud deployment frameworks and validation using larger industrial datasets.

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